

AM 463: QUIZ 8
Nov. 14, in class
15 minutes: Closed Book
Answer all problems
Presentation will affect your grade

Question 1:

i) State the viscous Burgers equation for $B(x, t)$.

ii) The travelling wave solution of this equation is given by

$$B(\xi) = B_u + \Delta B \frac{1}{1 + \exp[\Delta B \xi / 2\nu]}.$$

where $\xi = x - Vt$ and $\Delta B = B_d - B_u$. What are B_d and B_u ? Sketch a sample solution and one with a larger ΔB . How does the behaviour of the two differ?

iii) Consider the inviscid limit $\nu \rightarrow 0^+$. How does the solution change? What important fact does this tell us about shocks?